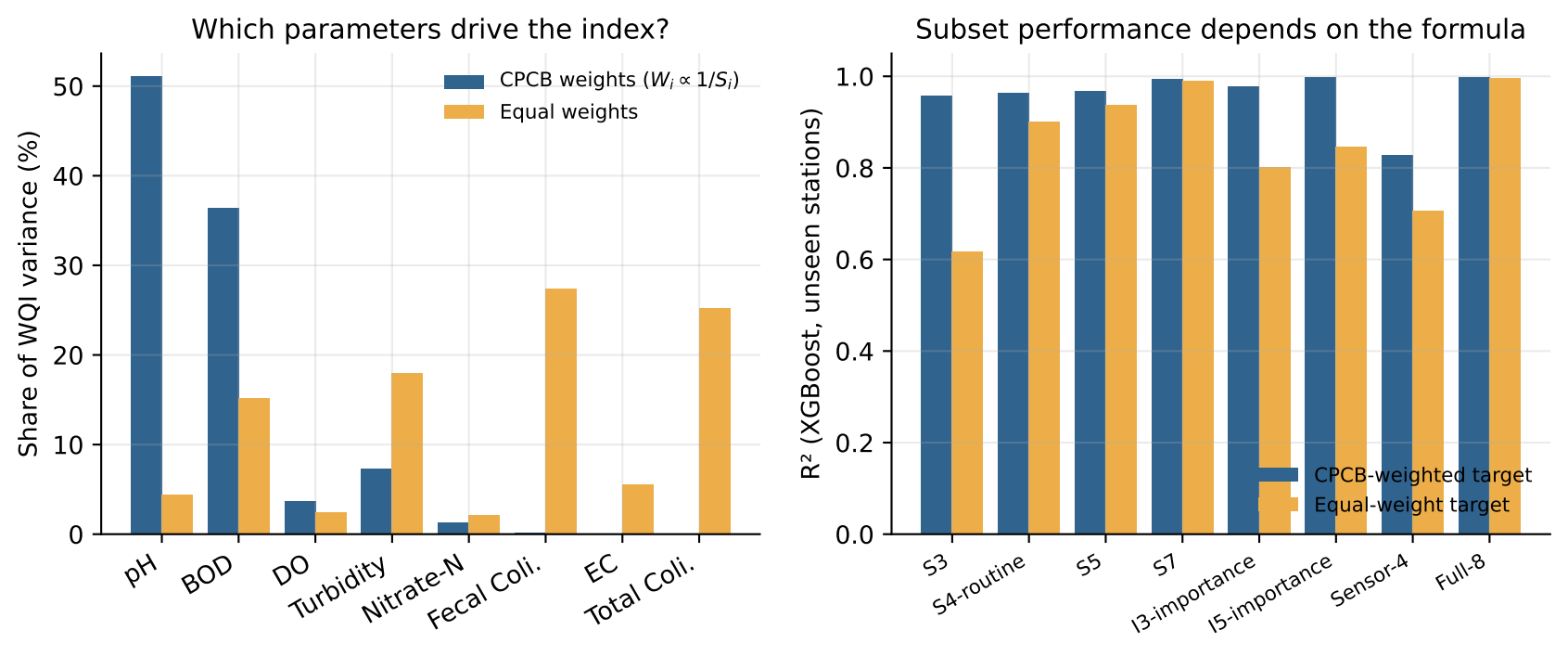
Which parameters drive the index?
Share of WQI variance (%)
CPCB weights ($W_i \propto 1/S_i$)
Equal weights
0
10
20
30
40
50
pH
BOD
DO
Turbidity
Nitrate-N
Fecal Coli.
EC
Total Coli.
Subset performance depends on the formula
R² (XGBoost, unseen stations)
0.0
0.2
0.4
0.6
0.8
1.0
CPCB-weighted target
Equal-weight target
S3
S4-routine
S5
S7
I3-importance
I5-importance
Sensor-4
Full-8